

Program : Diploma in Renewable Energy	
Course Code : 5299C	Course Title: Digital Electronics Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To familiarize with Digital IC trainer kit and verification of boolean expressions
- To design and implement various combinational logic circuits
- To design and implement Multiplexer and Decoder circuits
- To design and implement Sequential logic circuits

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge of Digital Electronics and Microprocessor		Digital Electronics and Microprocessor	5

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Build logic circuits using digital ICs	9	Applying
CO2	Design and implement Combinational logic circuits	9	Applying
CO3	Design and implement Multiplexer and Decoder circuits	9	Applying
CO4	Design and implement Sequential logic circuit	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Build logic circuits using digital ICs		
M1.01	Familiarization of Digital ICs and IC trainer kit	3	Understanding
M1.02	Verification and Realization of De Morgan's theorem	3	Applying
M1.03	Implement given logic functions using basic gates/universal gates	3	Applying
CO2	Design and implement Combinational logic circuits		
M2.01	Design, setup and verify the (i) Half and full adder circuits (ii) Half and full subtractor circuits	3	Applying
M2.02	Design and set up a Binary to gray and Gray to binary converter.	3	Applying
M2.03	Design and set up (1)Four bit adder /subtracter circuit using ICs 7483 (2) BCD adder using IC 7483.	3	Applying
	Lab Test – I	3	

CO3	Design and implement Multiplexer and Decoder circuits		
M3.01	Study of Multiplexer IC 74151 and Implement combinational logic using multiplexer ICs	3	Applying
M3.02	Design and set up BCD to Decimal decoder	3	Applying
M3.03	Verify the truth table SR, D, JK, T Flip flops using NAND gate/FF IC's	3	Applying
C04	Design and implement Sequential logic circuits		
M4.01	Design and set up 4 bit Serial In parallel out shift registers using flip flops	3	Applying
M4.02	Design and set up Ring counter and Johnsons counter using flip flops	3	Applying
M4.03	Design and set up Mode N Asynchronous up,down counters	3	Applying
M4.04	Design and set up Mode N Synchronous up , down counters	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	K.A Navas , Electronics Lab Manual - Vol 1 PHI Learning,ISBN : 9788120351424
T2	K.A Navas , Electronics Lab Manual - Vol 2,PHI Learning, ISBN : 9789388028080
T3	A Anand Kumar, Fundamental of Digital Electronics ,Prentice Hall

Online Resources

Sl.No	Website Link
1	https://www.youtube.com/watch?v=M6H49R87lyY
2	https://de-iitr.vlabs.ac.in/exp/4bit-synchronous-asynchronous-counter/theory.html
3	https://de-iitr.vlabs.ac.in/exp/multiplexer-demultiplexer/theory.html
4	https://www.youtube.com/watch?v=GwRw-yFUYJw
5	https://www.youtube.com/watch?v=5GEu1boOGOs
6	https://www.electrical4u.com/bcd-to-seven-segment-decoder/

Sample questions

1. Construct a Gray to Binary code converter
2. Construct a Serial in Parallel out shift register
3. Design a 3 bit synchronous up counter
4. Design and set up an UP/DOWN counter using MODE control switch
5. Seven segment Display

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open-ended experiment of their choice.